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COURSE: CSA 1613 – DWDM

CLASS TEST

CO_2 – AT2

Q1. a) DATA MINING AND DATA MINING TASK:

DATA MINING:

Data Mining is the process of extracting useful Patterns, Relationship and Knowledge from Large amount of data.

TWO DATA MINING TASK: i) classification
ii) Clustering

Steps in KDD Process:

1. Data cleaning
2. Data Integration
3. Data Selection
4. Data Transformation
5. Data Mining
6. Pattern Evalution

(b) DATA CLEANING AND HANDLING MISSING VALUES:

Data cleaning is the process of detecting and correcting inaccurate, incomplete, or inconsistent data.

TWO DATA QUALITY ISSUES: i) Missing Values
ii) Noisy data.

Handling Missing Values: i) Ignore The Record
ii) Replace with mean

Handling Noisy data: i) Binning
ii) Regression
iii) Clust

Q2. b) COVARIANCE ANALYSIS:

Covariance measures how Two Variable change Together.

Two Covariance Formulas:

Population Covariance:

$$Cov(x, y) = \frac{\sum (x_i - \mu x)(y_i - \mu y)}{N}$$

Sample Covariance:

$$Cov(x, y) = \frac{\sum (x_i - x)(y_i - y)}{n-1}$$

Q3 (a)

Given: Data (2,5), (3,8) (5,10)(4,11) (6,14)

$$\text{Mean of } x = \frac{2+3+5+4+6}{5} = \frac{20}{5} = 4$$

$$\text{Mean of } y = \frac{5+8+10+11+14}{5} = \frac{48}{5} = 9.6$$

Calculate deviations

x	y	$x-\bar{x}$	$y-\bar{y}$	Product
2	5	-2	-4.6	9.2
3	8	-1	-1.6	1.6
5	10	1	0.4	0.4
4	11	0	1.4	0
6	14	2	4.4	8.8

$$\text{Cov}(x,y) = \frac{\Sigma(x-x)(y-y)}{n}$$

$$= \frac{20}{5} \longrightarrow 9.2 + 1.6 + 0.4 + 0 + 8.8 = 20$$

$\text{Cov}(x,y) = 4$

$\text{Cov}(x,y) > 0$

$\therefore$ Stock A and Stock B Tend to Rise and fall Together.

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a) Given

$x = 4, 5, 6, 7, 9$

$y = 8, 3, 7, 5, 6$

$$\text{Mean } x = \frac{4+5+6+7+9}{5} = 6.2$$

$$\text{Mean } y = \frac{8+3+7+5+6}{5} = 5.8$$

Calculate deviation

x	y	x - 6.2	y - 5.8	$((x-6.2)(y-5.8))$
4	8	-2.2	2.2	-4.84
5	3	-1.2	-2.8	3.36
6	7	-0.2	1.2	-0.24
7	5	0.8	-0.8	-0.64
9	6	2.8	0.2	0.56

$$\Sigma(x-\bar{x})(y-\bar{y}) = -4.84 + 3.36 - 0.24 - 0.64 + 0.56 = -1.80$$

$$\text{Cov}(x,y) = \frac{\Sigma(x-x)(y-\bar{y})}{N} = \frac{-1.80}{5}$$

$\text{Cov}(x,y) = -0.36$

∴ Since the Covariance is Negative X and Y have a Negative Relationship. when X increases, Y Tends to decrease.

Q5 (a) Given:

Sorted Data: 13, 15, 16, 16, 19, 20, 20, 21, 22
22, 25, 25, 25, 30, 33, 33, 35
35, 35, 35, 36, 40, 45, 46, 52, 70

ToTal = 22, Divide into 3 equal frequency Bins

Each Bin contain 9 Values.

Bin 1 => 13, 15, 16, 16, 19, 20, 20, 21, 22

$$\text{Mean} = \frac{13+15+16+16+19+20+20+21+22}{9} = \frac{162}{9} = 18$$

Smoothed:

18, 18, 18, 18, 18, 18, 18, 18, 18

Bin 2 => 22, 25, 25, 25, 30, 33, 33, 35

$$\text{Mean} = \frac{22+25+25+25+30+33+33+35}{9} = \frac{253}{9} = 28.11$$

Smoothed:

28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11

Bin 3 => $\frac{35+35+35+36+40+45+46+52+70}{8}$

$$\text{Mean} = \frac{359}{8} = 44.88$$

Smoothed:

44.8, 44.8, 44.8, 44.8, 44.8, 44.8, 44.8, 44.8, 44.8